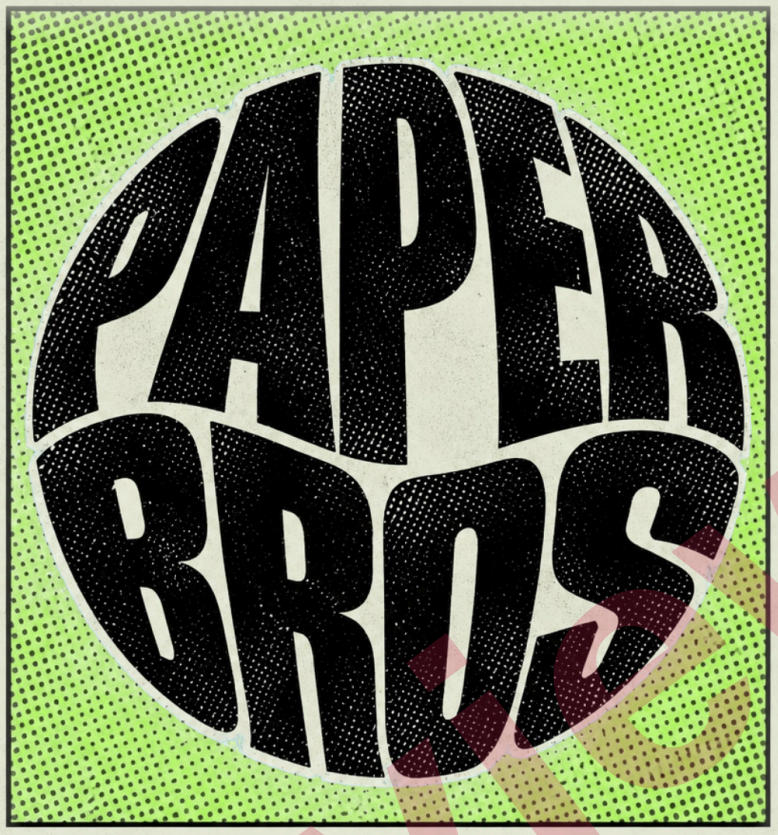
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A ONE STOP SOLUTION FOR YOUR BOARDS

DEVELOPED BY STUDENTS, FOR STUDENTS

CAMBRIDGE O LEVEL

MATHEMATICS (SYLLABUS D)

4024 · PAPER 2

TOPICAL QUESTIONS & MARK SCHEMES

MAY/JUNE 2015 – MAY/JUNE 2026

573 QUESTIONS · FULL WORKING SPACE · MARK SCHEMES FROM PAGE 762

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Mathematics 4024 · Paper 2 · Questions

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Topic 1: Number

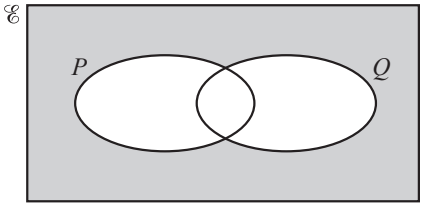
1.1 Types of number

8 entries

Q1 **2018/May-June/21/Q1 (c)**

earlier parts included for context (marked with a bar)

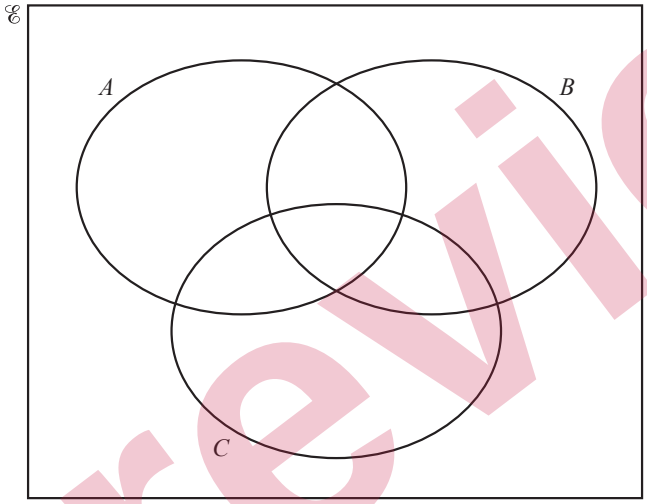
1 **(a)** Use set notation to describe the shaded region in the Venn diagram.



Answer ..... [1]

(b) $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$
$A = \{x : x \text{ is a factor of } 12\}$
$B = \{x : x \text{ is a multiple of } 2\}$
$C = \{x : x \text{ is a square number}\}$

(i) Show this information on the Venn diagram below.



[2]

(ii) Find $n(A \cap B)$.

Answer ..... [1]

(iii) Find $n(A \cap (B \cup C)')$.

Answer ..... [1]

(iv) One subset in the Venn diagram in **part (b)(i)** has no elements.

Use set notation to describe this subset.

Answer ..... [1]

(c) **(i)** Write 540 as the product of its prime factors.

Answer ..... [2]

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(ii) p is the smallest possible integer such that $540p$ is a square number.

Find $\sqrt{540p}$, giving your answer as the product of its prime factors.

Answer ..... [2]

Q2 2018/May-June/22/Q4 (b)(c)

(b) Find the lowest common multiple (LCM) of 140 and 770.

Answer ..... [2]

(c) A rectangular field measures 450 m by 306 m.
The whole field is divided into identical square plots with no land remaining.

Find the largest possible side length for the squares.

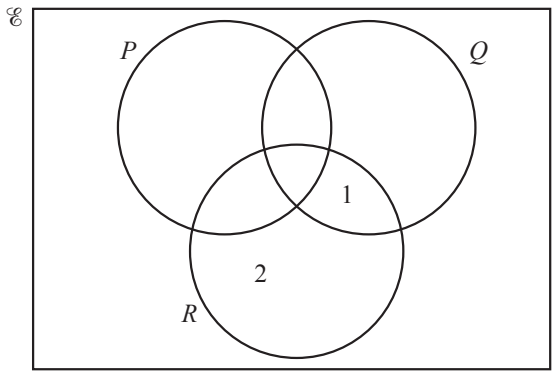
Answer ..... m [2]

Q3 2023/May-June/21/Q5 (b)

earlier parts included for context (marked with a bar)

5 **(a)** $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$
$P = \{x : x \text{ is a multiple of } 3\}$
$Q = \{x : x \text{ is an odd number}\}$
$R = \{x : x \text{ is a factor of } 24\}$

(i) Complete the Venn diagram.



[3]

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